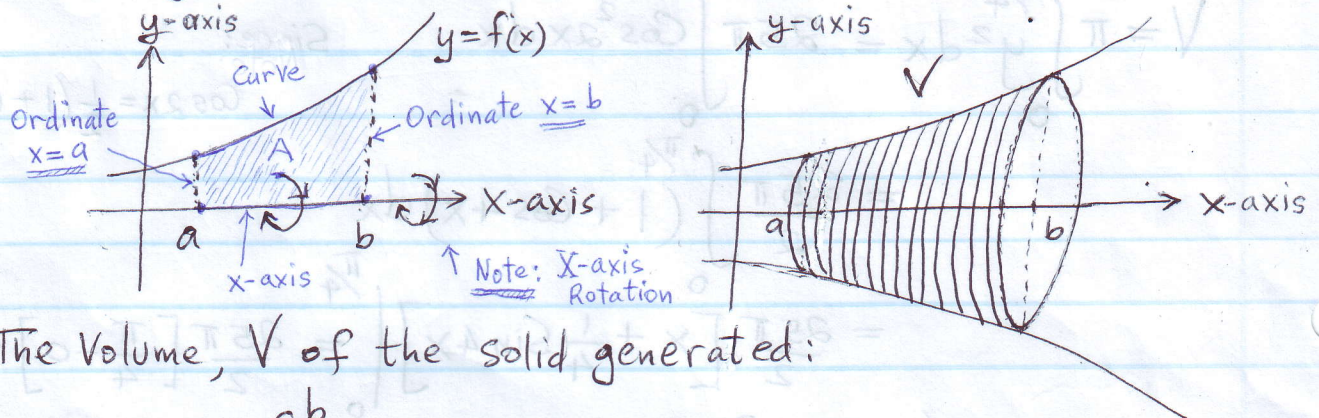


APPLICATION OF INTEGRATION

Lesson 13

1) VOLUME OF A SOLID OF REVOLUTION

If the curve $y=f(x)$, the x -axis and the ordinates at $x=a$ and $x=b$, rotates through a complete revolution about the x -axis, it will generate a solid of known Volume:

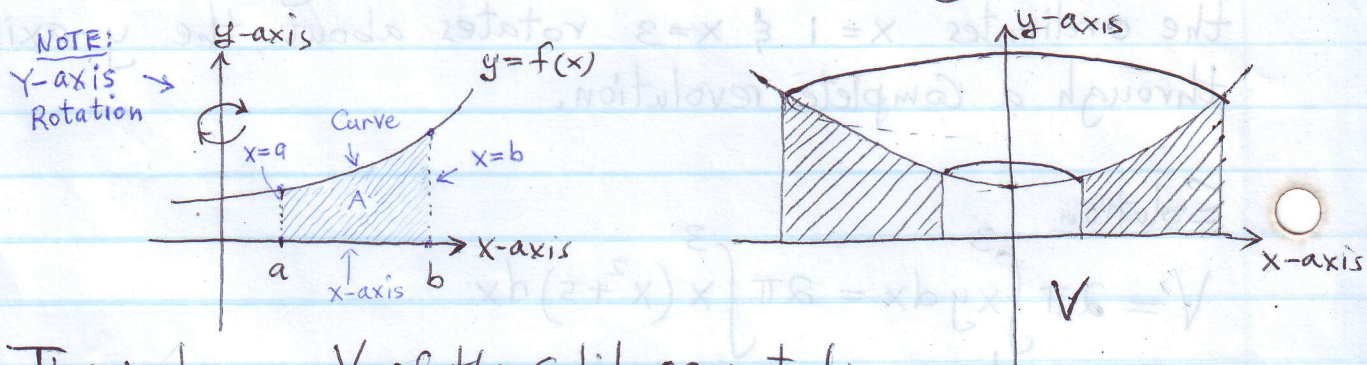


The Volume, V of the solid generated:

$$\textcircled{1} \quad V = \pi \int_a^b y^2 dx \quad (y=f(x) - \text{rotated abt } x\text{-axis}).$$

~~//////////~~

**If the curve $y=f(x)$ is rotated the y -axis, we have:



The volume, V of the solid generated:

$$\textcircled{2} \quad V = 2\pi \int_a^b xy dx \quad (y=f(x) - \text{rotated abt } y\text{-axis}).$$

~~//////////~~

* NOTE: The formulas $\textcircled{1}$ and $\textcircled{2}$ are sometimes called Shell formula for revolution about a horizontal and vertical lines.
(ie $\uparrow$ x -axis) (ie $\uparrow$ y -axis).

Example 1

Find the volume generated when the curve $y = 5 \cos 2x$, the x -axis and ordinates at $x=0$ & $x=\frac{\pi}{4}$ rotates about the x -axis through a complete revolution.

Solution:

$$\begin{aligned}
 V &= \pi \int_0^{\pi/4} y^2 dx = 25\pi \int_0^{\pi/4} \cos^2 2x dx \quad \text{Since:} \\
 & \qquad \qquad \qquad \cos 2x = \frac{1}{2}(1 + \cos 4x) \\
 &= \frac{25\pi}{2} \int_0^{\pi/4} (1 + \cos 4x) dx \\
 &= \frac{25\pi}{2} \left[x + \frac{1}{4} \sin 4x \right] \Big|_0^{\pi/4} = \frac{25\pi}{2} \left[\frac{\pi}{4} - 0 \right] = \frac{25\pi^2}{8} \\
 \therefore V &= \frac{25\pi^2}{8} \text{ units}^3
 \end{aligned}$$

Example 2

Find the volume generated when the curve $y = x^2 + 5$, the x -axis, the ordinates $x=1$ & $x=3$ rotates about the y -axis through a complete revolution.

Solution

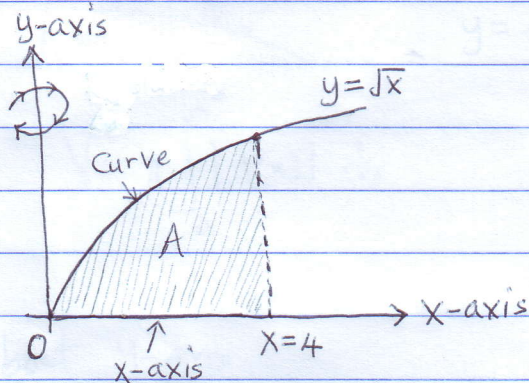
$$\begin{aligned}
 V &= 2\pi \int_1^3 xy dx = 2\pi \int_1^3 x(x^2 + 5) dx \\
 &= 2\pi \int_1^3 (x^3 + 5x) dx \\
 &= 2\pi \left[\frac{1}{4} x^4 + \frac{5}{2} x^2 \right] \Big|_1^3 = 2\pi \left[\frac{80}{4} + \frac{40}{2} \right] = 80\pi \\
 \therefore V &= 80\pi \text{ units}^3
 \end{aligned}$$

Example 3

The region bounded by the curve $y = \sqrt{x}$, the x-axis and the line (ordinate) $x=4$ is revolved about the y-axis to generate a solid. Find the volume of the solid.

Solution

* Since the curve is rotated about y-axis



$$V = 2\pi \int_0^4 xy \, dx$$

$$\text{but } y = \sqrt{x} = x^{1/2}$$

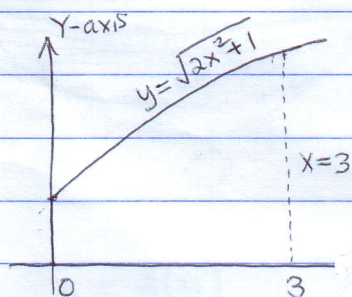
So that Volume:
$$V = 2\pi \int_0^4 x^{3/2} \, dx$$

$$= 2\pi \left[\frac{2x^{5/2}}{5} \right]_0^4 = \frac{4\pi}{5} \left[4^{5/2} - 0 \right] = \frac{4\pi}{5} (32) = \frac{128\pi}{5}$$

$$\therefore V = \frac{128\pi}{5} \text{ units}^3$$

Example 4

The area bounded by the curve $y = \sqrt{2x^2 + 1}$, the x-axis, the y-axis and the ordinate $x=3$ is revolved about the x-axis to generate a solid. Find the volume of the solid.

Solution

Rotation
about X-axis:

$$V = \pi \int_0^3 y^2 \, dx$$

$$= \pi \int_0^3 (2x^2 + 1) \, dx$$

$$= \pi \left[\frac{2x^3}{3} + x \right]_0^3 - \pi [0 + 0]$$

$$= \pi \left[\frac{2(3)^3}{3} + 3 \right]$$

$$\therefore V = 21\pi \text{ unit}^3$$